ALL STRUCTURE MAPS OF A HYPERCOVERING ARE COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1914-1923 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 1925-1942, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_hypercovering.json (pb=701, stmt 1914-1923, proof 1925-1942).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01G3: sources/stacks-hypercovering/pinned-original-source.tex, source lines 510–521. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01G5: sources/stacks-hypercovering/pinned-original-source.tex, source lines 563–575. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01GN: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1890–1904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01G4: sources/stacks-hypercovering/pinned-original-source.tex, source lines 524–535. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 017L: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1942–1942. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let K be a hypercovering of X. Then*

(1) *K_n is a covering of X for each $n \geq 0$,*
(2) *$d_i^n : K_n \to K_{n-1}$ is a covering for all $n \geq 1$ and $0 \leq i \leq n$.*

Proof. Recall that K_0 is a covering of X by Definition 01G5 and that this is equivalent to saying that $K_0 \to \{X \to X\}$ is a covering in the sense of Definition 01G3. Hence (1) follows from (2) because it will prove that the composition $K_n \to K_{n-1} \to \ldots \to K_0 \to \{X \to X\}$ is a covering by Lemma 01G4.

Proof of (2). Observe that $\mathrm{Mor}(\Delta[n], K)_0 = K_n$ by Simplicial, Lemma 017L. Therefore (2) follows from Lemma 01GN applied to the $n+1$ different inclusions $\Delta[n-1] \to \Delta[n]$. □